GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS
PHASE 2: Requirement Analysis

Project theme: historical Olympic participation and medal performance.

Objective

Define the user problem, intended audience, and what the Olympic analysis solution must provide.

Problem statement

Historical Olympic athlete-event data is detailed but difficult to summarize consistently across countries, sports, seasons, and years. Users need a clear and trustworthy way to explore participation and medal performance.

How might we

How might we transform historical Olympic athlete-event records into clear, verifiable comparisons of participation and medal performance across countries, sports, seasons, and years?

Target users

- Students and educators learning data analysis.
- Sports analysts and researchers comparing performance.
- Olympic fans and journalists exploring historical patterns.
- Sports administrators interested in high-level trends.

User needs

- Filter or compare records by year, season, NOC/team, sport, event, and sex.
- Distinguish medal counts from athlete-event participation.
- Understand missing values and dataset limitations.
- View concise summaries and readable charts.
- Export summary data for further analysis.

Functional requirements

- Load and validate the supplied CSV dataset.
- Clean and summarize the data by relevant dimensions.
- Calculate medal counts and participation measures.
- Create charts/tables and explain the results.
- Export analysis-ready data and document methodology.

Non-functional requirements

- Accuracy: calculations should be reproducible and checked.
- Usability: labels and charts should be understandable.
- Performance: use a reduced dataset under 25 MB when needed.
- Maintainability: document column meanings and processing steps.
- Transparency: describe missing data and limitations.

Expected phase deliverable

Problem statement, target users, and requirements.